Concept Review

Light Resistors

Why Do We Need Light Resistors?

Light resistors are simple, inexpensive sensors used for light detection when there is less need for accuracy. They are a good choice for systems that respond to large changes in light, such as automated lighting, smoke detectors, and security alarms.

Light Resistors

A light resistor, also called a light dependent resistor (LDR) or photoresistor, is a variable resistor that detects the amount of light on its surface. As the amount of light increases, its resistance decreases. A light resistor is shown in the image below:

Figure 1: Light resistor. Image source: https://www.pcboard.ca/ldr-light-dependent-resistor

Reading from Light Resistors

To measure from a light resistor, it can be connected in a voltage divider circuit. A voltage divider circuit is shown below:

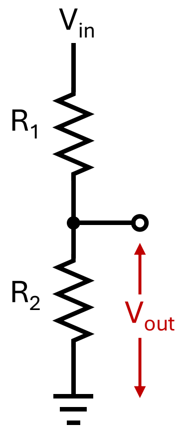

Figure 2: Voltage divider circuit

The output voltage $\mathrm{V_{out}}$ is a function of the input voltage V_{in} and the resistances $\mathrm{R_1}$ and $\mathrm{R_2}$:

$$\mathrm{V_{out}} = \left(\frac{\mathrm{R_2}}{\mathrm{R_1} + \mathrm{R_2}}\right) \times V_{\mathrm{in}}$$

For a voltage divider with a light resistor as $\mathrm{R_1}$ or $\mathrm{R_2}$, the output of the voltage divider changes with the resistance of the light resistor. The output voltage can be measured using the analog input pin of a microcontroller.

Datasheet and Considerations

Photoresistors are a simple, low-cost choice for applications that do not require complexity beyond detecting changes in light, such as automated lighting and line-following robots.

There are a few key specifications to consider when determining which light resistor is suitable for your application:

Dark resistance: This is the nominal resistance in total darkness.

Light resistance: This is the nominal resistance in bright light, measured under a specific amount of light in lux at a specific light temperature in Kelvins (K).

Response time: This refers to the time for the resistance to change in response to changes in light, which is typically in the range of tens of milliseconds.

Spectral peak: This is the specific wavelength of light that the light resistor detects with the most sensitivity. Light resistors detect changes within the visible light spectrum but differ in sensitivity depending on the wavelength of light.

The resistance response is also affected by temperature and light color temperature, which will vary depending on the environment. As such, light resistors are not suitable for high accuracy light measurement. For applications that require precise measurements and quick response time, photodiodes are usually selected instead of photoresistors.

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